

microeconomics and macroeconomics

- 1 Supply, Demand, Elasticity
 - supply and demand
 - revenue, cost, and profit
 - elasticity of demand
- 2 Duopolistic Competition and Production
 - duopolistic competition
 - theory of production
 - an input/output model
- 3 Proposals of Project Topics
 - Nash equilibria
 - model a price war
 - interval arithmetic to solve linear systems

MCS 472 Lecture 19
Industrial Math & Computation
Jan Verschelde, 25 February 2026

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the supply and demand functions

Consider a market of a single commodity with many producers and many consumers.

Definition (the demand function)

The *demand* function $q = D(p)$ gives the quantity q of units that will sell at price p .

Demand decreases with increasing price.

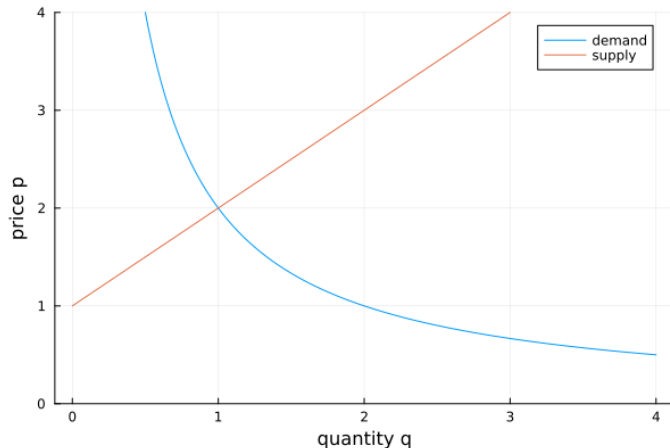
Definition (the supply function)

The *supply* function $q = S(p)$ gives the number of units that all producers will supply at price p .

Supply increases with increasing price.

a numerical example

Consider demand $D(p) = 2/p$ and supply $S(p) = p - 1$.

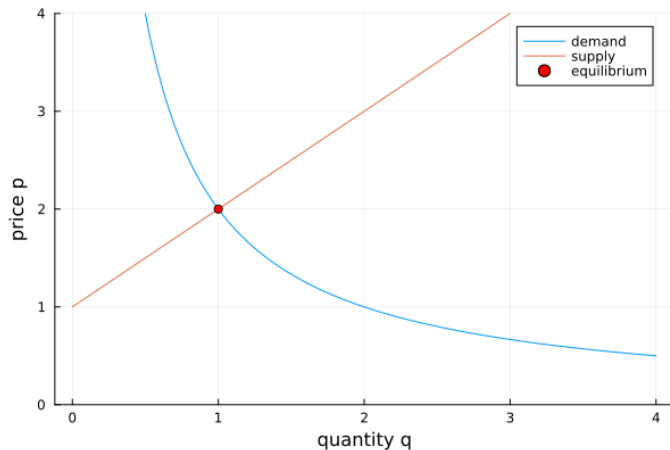


Traditionally, the quantity q appears on the horizontal axes, so we plot $D_q(q) = 2/q$ and $S_q(q) = 1 + q$.

the equilibrium price

Definition (the equilibrium price)

Given demand $D(p)$ and supply $S(p)$, the *equilibrium price* p_0 is the intersection of demand and supply: $D(p_0) = q_0 = S(p_0)$.



prices converge to the equilibrium

Prices naturally converge to the equilibrium.

- At prices above the equilibrium, more goods will be produced than the market can absorb and this *surplus* drives down prices.
- At prices below the equilibrium, when demand is higher than the supply, this *scarcity* will lift prices upwards.

the effect of a sales tax

What is the effect of a sales tax?

- Supply remains, because the consumer pays the tax.
- What is the new demand function?

The consumer pays $p(1 + T)$, where T is the tax, e.g.: $T = 10\%$.

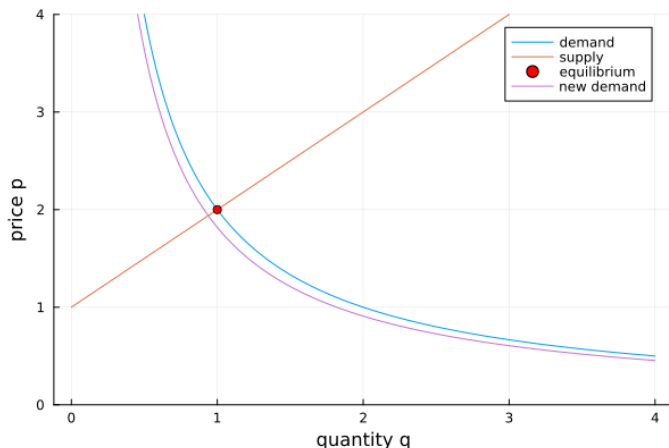
The new demand function is therefore

$$q = D_T(p) = D(p + Tp),$$

where $D(p)$ is the original demand, before the tax.

a sales tax of 10% depresses demand

For $D(p) = 2/p$, the new demand is $D(p + Tp) = 2/(p + Tp) = q$.



For plotting p in function of q : $D_q(q) = 2/(q(1 + T))$.

a tax on the producer

Exercise 1:

Consider $S(p) = 20p - 50$ and $D(p) = 100 - 30p$.

Suppose the producer must pay a specific tax of \$1 per unit.

- 1 What is the before tax equilibrium price?
- 2 What is the after tax equilibrium price?
- 3 How much of the tax is then actually paid by the consumer?

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revenue, cost, and profit

Definition (revenue)

The **revenue** of a producer $R(p, q)$ equals $p \cdot q$, the quantity q sold times the price p received for each unit sold.

At the equilibrium price p_0 , the revenue is $R(q) = p_0 q$.

Definition (cost)

The **cost** of a producer $C(q)$ is the cost to produce q units which includes the **fixed cost** equal to $C(0)$.

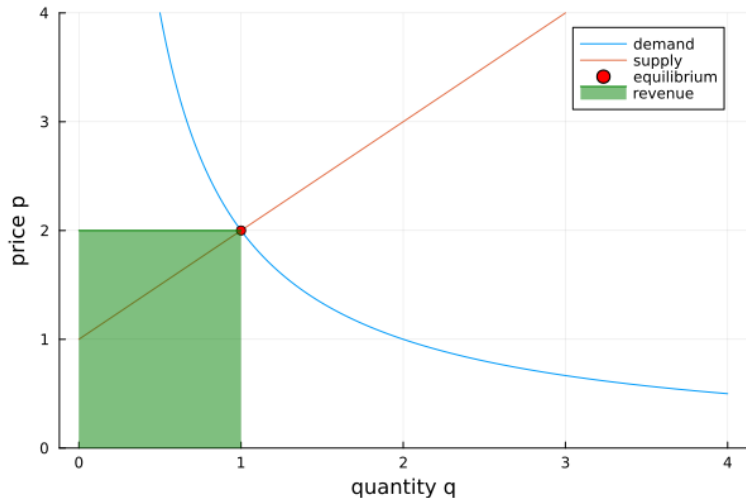
The cost function increases with q and tends to flatten.

Definition (profit)

The **profit** of a producer $P(q) = R(q) - C(q)$, revenue minus cost.

revenue as area

Revenue $R(p, q) = p \cdot q$.



effect of the sales tax on revenue

Recall the effect of a sales tax.

- 1 The sales tax depresses the demand.
- 2 The quantity sold decreases.
- 3 The equilibrium price drops.

Consequently, the revenue $R(p, q)$ decreases.

The *public sector revenue* from a sales tax T is

$$G(p, q) = T \cdot p \cdot q.$$

marginal cost and marginal revenue

Definition (marginal cost)

The *marginal cost* is the additional cost to produce one more unit.

For the cost function $C(q)$ we have

$$\frac{C(q+1) - C(q)}{1} \approx C'(q).$$

Definition (marginal revenue)

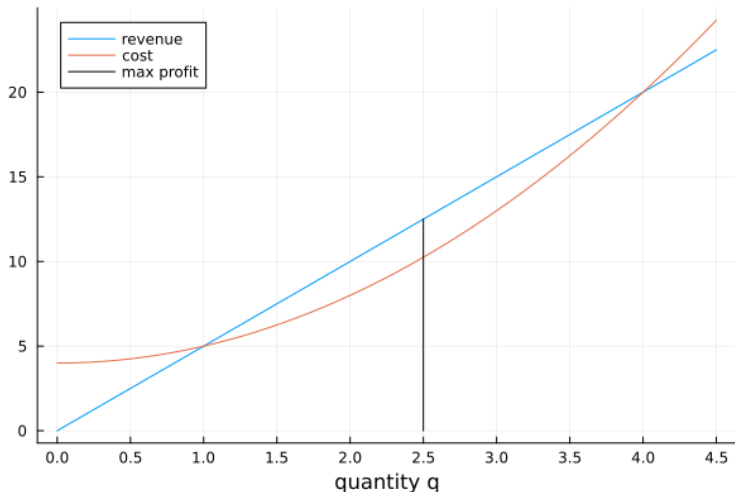
The *marginal revenue* is the additional revenue from one more unit.

Theorem (optimal profit)

Profit is optimal when $R'(q) = C'(q)$.

a numerical example

Revenue $R(q) = 5q$ while cost is $C(q) = 4 + q^2$.



Maximum profit when $R'(q) = 5 = 2q = C'(q)$, at $q = 2.5$.

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elasticity of demand

By how much can a producer raise the price of a product?

$$R(p, q) = p \cdot q, \quad \text{as } q = D(p), \quad \text{we have } R(p, q) = p \cdot D(p).$$

What is the change in demand as the price changes?

Definition (elasticity of demand)

The *elasticity of demand* e is

$$e = - \frac{\text{change in demand}}{\text{change in price}}.$$

The sign of e is negative as demand decreases for increasing price.

an expression for the elasticity

$$\begin{aligned}e &= - \frac{\text{change in demand}}{\text{change in price}} \\&= - \frac{dq/q}{dp/p} \\&= - \frac{dq}{dp} \cdot \frac{p}{q}, \quad q = D(p), \quad dq = D(p + dp) - D(p) \\&= -p \left(\frac{1}{D(p)} \right) \left(\frac{D(p + dp) - D(p)}{dp} \right)\end{aligned}$$

Theorem (elasticity as a function of price)

$$e = -p \frac{D'(p)}{D(p)}$$

revenue as a function of elasticity

What is the change in revenue as the price changes?

$$\begin{aligned}\frac{dR}{dp} &= \frac{d}{dp} \left(p \cdot D(p) \right), \quad \text{as } R(p, q) = p \cdot D(p) \\ &= 1 \cdot D(p) + p \cdot D'(p) \\ &= D(p) \left(1 + p \frac{D'(p)}{D(p)} \right), \quad e = -p \frac{D'(p)}{D(p)} \\ &= D(p)(1 - e)\end{aligned}$$

elastic and inelastic demand

The change in revenue in function of demand and elasticity:

$$\frac{dR}{dp} = D(p)(1 - e).$$

Case analysis:

- $e > 1$: revenue decreases when price increases.

We then say that the demand is *elastic*.

- $e < 1$: revenue increases when price increases.

We then say that the demand is *inelastic*.

- $e = 1$: implies $\frac{dR}{dp} = 0$, optimal revenue.

This analysis determines the price setting in a *monopoly*.

elasticity for a linear demand function

Exercise 2:

Consider a linear demand function, represented by $D(p) = q_0 - mp$.

- 1 Compute the elasticity e .
- 2 For which price p is $e = 1$?

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duopolistic competition

In microeconomics, we consider only one producer.

Now we consider two competing producers, going to a larger scale, into macroeconomics.

We make two assumptions.

- 1 The demand function is the same for both producers.
For example, bottled water.
- 2 Each producer believes the other will hold its output constant.
This model was proposed by Cournot in 1838.

What is the price setting mechanism?

price dynamics

The competitors A and B have the same demand $q = D(p)$.

- 1 Assume A is alone on the market at first.

A sets the price according to elasticity $e = 1$:

$$p = a_1 = -\frac{D(a_1)}{D'(a_1)}.$$

- 2 B enters, the demand is decreased with $q_A = D(a_1)$.

The demand function for B is then $D(p) - D(a_1)$.

B sets the price according to elasticity $e = 1$:

$$p = b_1 = -\frac{D(b_1) - D(a_1)}{D'(b_1)}.$$

deriving a discrete time system

Rewrite

$$b_1 = -\frac{D(b_1) - D(a_1)}{D'(b_1)}$$

as

$$D'(b_1)b_1 + D(b_1) = D(a_1).$$

Now A reacts to the new demand curve $D(p) - D(b_1)$:

$$a_2 = -\frac{D(a_2) - D(b_1)}{D'(a_2)},$$

which can be rewritten as

$$D'(a_2)a_2 + D(a_2) = D(b_1).$$

a discrete time system

Assuming A is leader and B is follower, we start with

$$a_1 = -\frac{D(a_1)}{D'(a_1)},$$

and continue for $k = 1, 2 \dots$ with

$$D'(b_k)b_k + D(b_k) = D(a_k)$$

and

$$D'(a_{k+1})a_{k+1} + D(a_{k+1}) = D(b_k).$$

This can be formulated as a fixed-point iteration, converging to an equilibrium if the equations are contractive.

the model for a linear demand function

Exercise 3:

Consider the duopolistic competition for a linear demand function, represented by $D(p) = q_0 - mp$.

- 1 Verify that the model converges.
- 2 What is the equilibrium price?

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the Cobb-Douglas model

The Cobb-Douglas model expresses production Q as a function of

- L , labor, and
- K , capital investment

where A is some constant depending on technology:

$$Q = A L^{\alpha} K^{1-\alpha}, \quad 0 < \alpha < 1.$$

Note that $Q = Q(L, K)$ is homogeneous. For some $\lambda > 1$:

$$\begin{aligned} Q(\lambda L, \lambda K) &= A (\lambda L)^{\alpha} (\lambda K)^{1-\alpha} \\ &= A \lambda^{\alpha} L^{\alpha} \lambda^{1-\alpha} K^{1-\alpha} \\ &= A \lambda L^{\alpha} K^{1-\alpha} \\ &= \lambda Q(L, K) \end{aligned}$$

For example: doubling labor and capital doubles the output.

marginal product of labor

Production Q in function of labor L and capital K :

$$Q = A L^{\alpha} K^{1-\alpha}, \quad 0 < \alpha < 1.$$

What is the effect of the addition of one extra unit of labor?

Take the derivative of Q with respect to L :

$$w = \frac{dQ}{dL} = A \alpha L^{\alpha-1} K^{1-\alpha} = \alpha \frac{Q}{L}$$

is the *marginal product of labor*,

which should equal the wage w paid to a worker.

marginal product of capital

Production Q in function of labor L and capital K :

$$Q = A L^{\alpha} K^{1-\alpha}, \quad 0 < \alpha < 1.$$

What is the effect of the addition of one extra unit of capital?

Take the derivative of Q with respect to K :

$$r = \frac{dQ}{dK} = A(1 - \alpha)L^{\alpha}K^{1-\alpha-1} = (1 - \alpha)\frac{Q}{K}$$

is the *marginal product of capital*,
which should equal the rent r for capital.

a macroeconomic law

Production Q in function of labor L and capital K :

$$Q = A L^{\alpha} K^{1-\alpha}, \quad 0 < \alpha < 1, \quad w = \alpha \frac{Q}{L}, \quad r = (1 - \alpha) \frac{Q}{K}.$$

Observe

$$Lw + Kr = \alpha Q + (1 - \alpha)Q = Q,$$

independent of the technology A . So we derived:

Theorem (Cobb-Douglas model)

$$Q = wL + rK$$

Production is the sum of

- 1 marginal product of labor times labor, and
- 2 marginal product of capital time capital investment.

a microeconomic application

For a specific product made by a producer, determine in

$$Q = A L^{\alpha} K^{1-\alpha}, \quad 0 < \alpha < 1.$$

the parameters A and α by sampling and fitting.

Then consider

$$Q = wL + rK.$$

To increase production,

- 1 hire more workers if $w > r$, or
- 2 raise more capital if $w < r$.

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an input/output model

The Leontiev input/output model considers three sectors:

A : Agriculture I : Industry S : Service,

each producing x_1 , x_2 , x_3 goods respectively.

C is the demand of the Consumer sector.

The equation

$$x_1 = 0.3x_1 + 0.2x_2 + 0.3x_3 + 4$$

expresses that for all goods produced by Agriculture

- ① 30% is consumed by Agriculture,
- ② 20% is consumed by Industry,
- ③ 30% is consumed by Service,

and the remaining 4 units are absorbed by the Consumer sector.

a matrix relation

The three equations

$$x_1 = 0.3x_1 + 0.2x_2 + 0.3x_3 + 4$$

$$x_2 = 0.2x_1 + 0.4x_2 + 0.3x_3 + 5$$

$$x_3 = 0.2x_1 + 0.5x_2 + 0.1x_3 + 12$$

can be summarized into a single matrix relation

$$\mathbf{x} = A\mathbf{x} + \mathbf{b}.$$

The vector $\mathbf{b}$ is called *the bill of goods*.

Is the bill of goods realizable?

Is the bill of goods realizable?

The question is equivalent to asking whether

$$\mathbf{x} = A\mathbf{x} + \mathbf{b}$$

has a solution with all entries in $\mathbf{x}$ positive.

The solution is $\mathbf{x} = (I - A)^{-1}\mathbf{b}$.

The series expansion of $(I - A)^{-1}$ is

$$\frac{1}{I - A} = I + A + A^2 + \dots$$

which converges only if $\|A\|_2 < 1$,

if the dominant eigenvalue is less than one in modulus.

computations with Julia

```
julia> A = [0.3 0.2 0.3; 0.2 0.4 0.3; 0.2 0.5 0.1];
julia> b = [4; 5; 12];
julia> using LinearAlgebra
julia> I = diagm(ones(3));
julia> x = (I - A)\b
3-element Vector{Float64}:
 38.30188679245284
 44.33962264150945
 46.47798742138366
julia> L, v = eigen(A); L
3-element Vector{Float64}:
-0.17384200332092611
 0.12943335560447009
 0.8444086477164554
```

We see that the bill of goods is realizable.

interpretation of the solution

Exercise 4:

As the solution to the example, we obtained

$$x_1 \approx 38.30, \quad x_2 \approx 44.34, \quad x_3 \approx 46.48.$$

- 1 Write an interpretation for those numbers.
- 2 The largest eigenvalue is $0.84 < 1$. Modify the coefficients of the matrix so the largest eigenvalue increases.

What does an increase mean for the solution?

summary and bibliography

We considered mathematical models of markets, to study price setting mechanisms, in monopoly and duopoly.

This lecture follows Chapter 8 of our text book.

- Charles R. MacCluer:
Industrial Mathematics. Modeling in Industry, Science, and Government. Prentice Hall, 2000.

Available through our UIC library are online versions.

- Hans Peters: *Game Theory. A Multi-Leveled Approach*. Springer 2008. Page 7 gives an example of a Cournot game.
- Richard H. Enns: *It's a Nonlinear World*. Springer Undergraduate Texts in Mathematics and Technology, 2011. Chapter 11 describes nonlinear models of competition.

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1. Nash equilibria

The concept of a Nash equilibrium is important in game theory.

Read the Game Theory book of Hans Peters for the definition and provide a computational description of a good example.

Consider the following questions.

- 1 Formulate an economically interesting model.
- 2 Give an example of an application with many Nash equilibria.

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2. model a price war

Develop a computational model of a price war.

Read chapter 11 of *It's a Nonlinear World* by Richard Enns.

Consider the following questions.

- 1 Formulate a model with a plausible demand function.
- 2 Is it possible that chaos occurs?

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3. interval arithmetic to solve linear systems

Data from real applications often come with noise.

Read the paper **A method for solving systems of linear interval equations applied to the Leontief input-output of economics**, by L. Dymova, P. Sevastjanov, and M. Pilarek, *Expert Systems with Applications*, vol. 40, pages 222–230, 2013.

Consider the following:

- 1 Summarize the content of the paper, with special attention to the input-output model of economics.
- 2 Reproduce the computations on the examples in the paper, for example, using SageMath.